

2. *Cash or collateral* return

The cash return is the interest earned on the collateral (or the margin) investors are required to post to trade futures contracts.

3. *Roll* return

Investors desiring constant exposures to commodities must roll over their exposures to new futures contracts as the old ones expire. As they sell futures contracts that are about to expire and buy new ones, they incur capital gains or losses.

Table 11.7 breaks the average total returns of the GSCI into spot, cash, and roll return components. Figure 11.8 plots the rolling two-year average of these return components over time. From January 1970 to December 2011, the average 11.3% GSCI total return included a spot return of 6.4%, a cash return of 5.6% and a -0.7% roll return. During the 1980s, however, the spot return was negative, at -0.4% , but overall commodity futures returns were still positive because the losses in commodity spot markets were offset by high interest earnings on collateral, with a cash return of 9.1% , and a positive roll return of 2.4% . In the 2000s, the roll return was -7.5% , but spot commodity markets did well, at 13% , so the total GSCI return was positive.

Figure 11.8 shows that the variation in the spot return is much larger than the variation in the roll and cash returns. We can decompose the variation of total GSCI returns into:

$$\begin{aligned} \text{var}(\text{total return}) &= \text{var}(\text{spot}) + \text{var}(\text{cash}) + \text{var}(\text{roll}) \\ 100\% &\qquad 100.0\% \quad 0.2\% \quad 6.0\% \\ &+ 2\text{cov}(\text{spot}, \text{roll}) + 2\text{cov}(\text{spot}, \text{cash}) + 2\text{cov}(\text{roll}, \text{cash}) \\ &\qquad 2 \times -3.4\% \qquad 2 \times 0.2\% \qquad 2 \times 0.1\% \end{aligned} \tag{11.3}$$

Table 11.7

Average Returns of GSCI Commodities Index				
	Spot	Cash	Roll	Total
1970s	11.1%	6.6%	4.0%	21.6%
1980s	-0.4%	9.1%	2.4%	11.1%
1990s	0.8%	5.0%	-0.5%	5.3%
2000–2011	13.0%	2.3%	-7.5%	7.8%
1970–2011	6.4%	5.6%	-0.7%	11.3%